

Chapter index

Appendices · generated

Every chapter, with the file it came from

Use this when you want to go the other way — from something you remember reading here, back to the markdown (or the deck, or the code) that lives in the repository. **Ch.** is the chapter number used in the contents page and in the PDF outline; **Page** is where that chapter starts in this book.

Ch.	Title	Page	Source file
Part 0 — 🚀 Launchpad			
0.1	Overview	9	00-launchpad/README.md
Part 01 — 🏠 Machine Learning Foundations			
01.1	Overview	12	01-ml-foundations/README.md
01.2	Core Concepts — Machine Learning Foundations	14	01-ml-foundations/notes/01-core-concepts.md
01.3	Deep Dive — Optimization & Generalization	19	01-ml-foundations/notes/02-optimization-and-generalization.md
01.4	Lab exercises	23	01-ml-foundations/lab/EXERCISES.md
01.5	Primary sources & further reading	25	01-ml-foundations/papers/PAPERS.md 01-ml-foundations/resources/RESOURCES.md
Part 02 — 🧠 Deep Learning			
02.1	Overview	31	02-deep-learning/README.md
02.2	Core Concepts — Neural Networks	33	02-deep-learning/notes/01-neural-networks.md
02.3	Deep Dive — Training Deep Networks (the craft)	38	02-deep-learning/notes/02-training-deep-networks.md
02.4	Lab exercises	42	02-deep-learning/lab/EXERCISES.md
02.5	Primary sources & further reading	45	02-deep-learning/papers/PAPERS.md 02-deep-learning/resources/RESOURCES.md
Part 03 — 🗣️ Natural Language Processing			
03.1	Overview	50	03-nlp/README.md
03.2	Core Concepts — From Text to Vectors	52	03-nlp/notes/01-text-to-vectors.md
03.3	Deep Dive — The Transformer	56	03-nlp/notes/02-transformers.md
03.4	Lab exercises	62	03-nlp/lab/EXERCISES.md
03.5	Primary sources & further reading	64	03-nlp/papers/PAPERS.md 03-nlp/resources/RESOURCES.md
Part 04 — 📖 Large Language Models			

Ch.	Title	Page	Source file
04.1	Overview	69	04-llm/README.md
04.2	Core Concepts — How an LLM Is Built	71	04-llm/notes/01-how-llms-are-built.md
04.3	Deep Dive — Inference, Decoding & Adaptation	76	04-llm/notes/02-inference-and-adaptation.md
04.4	Lab exercises	80	04-llm/lab/EXERCISES.md
04.5	Code templates & reference notes	83	04-llm/code/prompt_patterns.md
04.6	Primary sources & further reading	87	04-llm/papers/PAPERS.md 04-llm/resources/RESOURCES.md
Part 05 — 🧠 Generative AI			
05.1	Overview	92	05-genai/README.md
05.2	Core Concepts — Diffusion and the Generative Landscape	94	05-genai/notes/01-core-concepts.md
05.3	Deep Dive — Pipelines, Multimodality, and Honest Evaluation	99	05-genai/notes/02-multimodal-and-practice.md
05.4	Lab exercises	104	05-genai/lab/EXERCISES.md
05.5	Primary sources & further reading	107	05-genai/papers/PAPERS.md 05-genai/resources/RESOURCES.md
Part 06 — 🤖 AI Agents			
06.1	Overview	113	06-ai-agents/README.md
06.2	Core Concepts — What Makes Something an Agent	115	06-ai-agents/notes/01-core-concepts.md
06.3	Deep Dive — Planning, Tool Design, Failure Modes, Evals, Security	122	06-ai-agents/notes/02-planning-failure-and-evals.md
06.4	Lab exercises	128	06-ai-agents/lab/EXERCISES.md
06.5	Primary sources & further reading	131	06-ai-agents/papers/PAPERS.md 06-ai-agents/resources/RESOURCES.md
Part 07 — 🌐 Agentic AI Systems			
07.1	Overview	137	07-agentic-ai/README.md
07.2	Core Concepts — The Autonomy Spectrum and Multi-Agent Topologies	139	07-agentic-ai/notes/01-autonomy-spectrum-and-topologies.md
07.3	Deep Dive — Cost, Protocols, Failure Modes, Control, and the Decision Framework	144	07-agentic-ai/notes/02-coordination-control-and-evaluation.md
07.4	Lab exercises	152	07-agentic-ai/lab/EXERCISES.md
07.5	Primary sources & further reading	155	07-agentic-ai/papers/PAPERS.md 07-agentic-ai/resources/RESOURCES.md
Part 08 — 🔍 Retrieval-Augmented Generation			
08.1	Overview	159	08-rag/README.md

Ch.	Title	Page	Source file
08.2	Core Concepts — Why RAG, Chunking, Embeddings, Vector Search	161	08-rag/notes/01-core-concepts.md
08.3	Deep Dive — Hybrid Search, Advanced Patterns, Generation, Evaluation, Failure Modes	167	08-rag/notes/02-advanced-patterns-and-evaluation.md
08.4	Lab exercises	174	08-rag/lab/EXERCISES.md
08.5	Primary sources & further reading	177	08-rag/papers/PAPERS.md 08-rag/resources/RESOURCES.md
Part 09 — 🦾 Model Context Protocol			
09.1	Overview	183	09-mcp/README.md
09.2	MCP Core Concepts	186	09-mcp/notes/01-core-concepts.md
09.3	MCP Security Model, Ecosystem, and Alternatives	192	09-mcp/notes/02-security-ecosystem-and-alternatives.md
09.4	Lab exercises	197	09-mcp/lab/EXERCISES.md
09.5	Primary sources & further reading	200	09-mcp/papers/PAPERS.md 09-mcp/resources/RESOURCES.md
Part 10 — 🏗️ AI Architecture			
10.1	Overview	205	10-ai-architecture/README.md
10.2	Core Concepts — The Layered Reference Architecture	207	10-ai-architecture/notes/01-reference-architecture.md
10.3	Deep Dive — Reliability, Observability, Security, Deployment, Cost, Capstone	215	10-ai-architecture/notes/02-production-patterns.md
10.4	Lab exercises	224	10-ai-architecture/lab/EXERCISES.md
10.5	Primary sources & further reading	226	10-ai-architecture/papers/PAPERS.md 10-ai-architecture/resources/RESOURCES.md
Part 11 — 🌐 The LLM Model Landscape			
11.1	Overview	230	11-llm-models/README.md
11.2	Core Concepts — The Model Landscape	232	11-llm-models/notes/01-the-model-landscape.md
11.3	Deep Dive — Selecting, Migrating, and Maintaining a Model Choice	238	11-llm-models/notes/02-selecting-and-migrating-models.md
11.4	Lab exercises	244	11-llm-models/lab/EXERCISES.md
11.5	Primary sources & further reading	247	11-llm-models/papers/PAPERS.md 11-llm-models/resources/RESOURCES.md
Part 12 — ⚖️ AI Governance			
12.1	Overview	251	12-ai-governance/README.md
12.2	Core Concepts — Why AI Governance, and the Frameworks That Structure It	253	12-ai-governance/notes/01-governance-fundamentals-and-frameworks.md

Ch.	Title	Page	Source file
12.3	Deep Dive — Operating Model, Oversight, Vendors, Change & Incident Practice	259	12-ai-governance/notes/02-operating-model-and-practice.md
12.4	Lab exercises	266	12-ai-governance/lab/EXERCISES.md
12.5	Code templates & reference notes	269	12-ai-governance/code/model_card_template.md
12.6	Primary sources & further reading	276	12-ai-governance/papers/PAPERS.md 12-ai-governance/resources/RESOURCES.md
Part 13 —  AI Security			
13.1	Overview	283	13-ai-security/README.md
13.2	Core Concepts — The OWASP Frame and Prompt Injection	285	13-ai-security/notes/01-owasp-and-prompt-injection.md
13.3	Deep Dive — Agentic Security, Data Risks, Red-Teaming, and the Supply Chain	292	13-ai-security/notes/02-agentic-security-and-red-teaming.md
13.4	Lab exercises	300	13-ai-security/lab/EXERCISES.md
13.5	Primary sources & further reading	304	13-ai-security/papers/PAPERS.md 13-ai-security/resources/RESOURCES.md
Part 14 —  AI Compliance			
14.1	Overview	310	14-ai-compliance/README.md
14.2	Core Concepts — The Regulatory Landscape	312	14-ai-compliance/notes/01-regulatory-landscape.md
14.3	Deep Dive — Sectors, Data Protection, Standards, and IP	318	14-ai-compliance/notes/02-sectors-data-protection-and-standards.md
14.4	Practitioner Craft — Building a Compliance Program	323	14-ai-compliance/notes/03-building-a-compliance-program.md
14.5	Lab exercises	327	14-ai-compliance/lab/EXERCISES.md
14.6	Primary sources & further reading	330	14-ai-compliance/papers/PAPERS.md 14-ai-compliance/resources/RESOURCES.md
Part 15 —  AI Evaluation			
15.1	Overview	337	15-ai-evals/README.md
15.2	Core Concepts — Why LLM Evaluation Is a Different Discipline	340	15-ai-evals/notes/01-core-concepts.md
15.3	Deep Dive — Judge Craft, Statistical Rigor, and Eval-Driven Development	345	15-ai-evals/notes/02-judge-craft-and-eval-driven-development.md
15.4	Lab exercises	352	15-ai-evals/lab/EXERCISES.md
15.5	Primary sources & further reading	355	15-ai-evals/papers/PAPERS.md 15-ai-evals/resources/RESOURCES.md
Part 16 —  AI Tech Stack			
16.0	Overview	361	16-ai-tech-stack/README.md
Track 16.1 —  Python for AI Engineering			

Ch.	Title	Page	Source file
16.1.1	Overview	364	16-ai-tech-stack/tracks/python/README.md
16.1.2	Core Concepts — The Modern Python Toolchain	366	16-ai-tech-stack/tracks/python/notes/01-modern-python-toolchain.md
16.1.3	Practitioner Craft — Numerics, Async, and the Idioms That Bite	374	16-ai-tech-stack/tracks/python/notes/02-numerics-async-and-performance.md
16.1.4	Lab exercises	383	16-ai-tech-stack/tracks/python/lab/EXERCISES.md
16.1.5	Primary sources & further reading	388	16-ai-tech-stack/tracks/python/papers/PAPERS.md 16-ai-tech-stack/tracks/python/resources/RESOURCES.md
Track 16.2 —  TypeScript for AI Applications			
16.2.1	Overview	396	16-ai-tech-stack/tracks/typescript/README.md
16.2.2	Core Concepts — TypeScript for AI Applications	398	16-ai-tech-stack/tracks/typescript/notes/01-core-concepts.md
16.2.3	Streaming, SDKs, Runtimes and Hygiene	405	16-ai-tech-stack/tracks/typescript/notes/02-streaming-runtimes-and-hygiene.md
16.2.4	Lab exercises	413	16-ai-tech-stack/tracks/typescript/lab/EXERCISES.md

Ch.	Title	Page	Source file
16.2.5	Code templates & reference notes	416	16-ai-tech-stack/tracks/typescript/code/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/@esbuild/darwin-arm64/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/@types/node/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/@typescript/typescript-darwin-arm64/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/esbuild/LICENSE.md 16-ai-tech-stack/tracks/typescript/code/node_modules/esbuild/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/tsx/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/typescript/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/typescript/vendor/vscode-jsonrpc/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/undici-types/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/zod/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/zod-to-json-schema/README.md 16-ai-tech-stack/tracks/typescript/code/node_modules/zod-to-json-schema/changelog.md 16-ai-tech-stack/tracks/typescript/code/node_modules/zod-to-json-schema/contributing.md
16.2.6	Primary sources & further reading	444	16-ai-tech-stack/tracks/typescript/papers/PAPERS.md 16-ai-tech-stack/tracks/typescript/resources/RESOURCES.md
Track 16.3 —  LangChain			
16.3.1	Overview	451	16-ai-tech-stack/tracks/langchain/README.md
16.3.2	Core Concepts — LangChain	453	16-ai-tech-stack/tracks/langchain/notes/01-core-concepts.md
16.3.3	Practitioner Craft — LangChain in Production	461	16-ai-tech-stack/tracks/langchain/notes/02-practitioner-craft.md
16.3.4	Lab exercises	467	16-ai-tech-stack/tracks/langchain/lab/EXERCISES.md

Ch.	Title	Page	Source file
16.3.5	Primary sources & further reading	471	16-ai-tech-stack/tracks/langchain/papers/PAPERS.md 16-ai-tech-stack/tracks/langchain/resources/RESOURCES.md
Track 16.4 — 📊 LangGraph			
16.4.1	Overview	477	16-ai-tech-stack/tracks/langgraph/README.md
16.4.2	Core Concepts — LangGraph	479	16-ai-tech-stack/tracks/langgraph/notes/01-core-concepts.md
16.4.3	Production Craft — Persistence, Approval, Durability, Multi-Agent	485	16-ai-tech-stack/tracks/langgraph/notes/02-production-craft.md
16.4.4	Lab exercises	492	16-ai-tech-stack/tracks/langgraph/lab/EXERCISES.md
16.4.5	Primary sources & further reading	496	16-ai-tech-stack/tracks/langgraph/papers/PAPERS.md 16-ai-tech-stack/tracks/langgraph/resources/RESOURCES.md
Track 16.5 — 📊 LangSmith			
16.5.1	Overview	501	16-ai-tech-stack/tracks/langsmith/README.md
16.5.2	Core Concepts — Tracing & the LangSmith Data Model	503	16-ai-tech-stack/tracks/langsmith/notes/01-core-concepts.md
16.5.3	Evaluation & Operations — Turning Traces Into a Ratchet	510	16-ai-tech-stack/tracks/langsmith/notes/02-evaluation-and-operations.md
16.5.4	Lab exercises	518	16-ai-tech-stack/tracks/langsmith/lab/EXERCISES.md
16.5.5	Primary sources & further reading	522	16-ai-tech-stack/tracks/langsmith/papers/PAPERS.md 16-ai-tech-stack/tracks/langsmith/resources/RESOURCES.md
Track 16.6 — ☁️ AI on AWS with Amazon Bedrock			
16.6.1	Overview	528	16-ai-tech-stack/tracks/aws-bedrock/README.md
16.6.2	Core Concepts — AI on AWS with Amazon Bedrock	530	16-ai-tech-stack/tracks/aws-bedrock/notes/01-core-concepts.md
16.6.3	Production Craft — Cost, Quotas, Security and the Architecture	541	16-ai-tech-stack/tracks/aws-bedrock/notes/02-production-craft.md
16.6.4	Lab exercises	550	16-ai-tech-stack/tracks/aws-bedrock/lab/EXERCISES.md
16.6.5	Code templates & reference notes	554	16-ai-tech-stack/tracks/aws-bedrock/code/iam_and_guardrails_examples.md
16.6.6	Primary sources & further reading	563	16-ai-tech-stack/tracks/aws-bedrock/papers/PAPERS.md 16-ai-tech-stack/tracks/aws-bedrock/resources/RESOURCES.md
Track 16.7 — 🖥️ Claude Code and the Agent SDK			
16.7.1	Overview	571	16-ai-tech-stack/tracks/claude-code/README.md

Ch.	Title	Page	Source file
16.7.2	Core Concepts — Claude Code as a Tool You Use Well	574	16-ai-tech-stack/tracks/claude-code/notes/01-claude-code-core-concepts.md
16.7.3	The Claude Agent SDK — Building On the Loop Instead of Rebuilding It	585	16-ai-tech-stack/tracks/claude-code/notes/02-agent-sdk-and-building-on-it.md
16.7.4	Lab exercises	594	16-ai-tech-stack/tracks/claude-code/lab/EXERCISES.md
16.7.5	Code templates & reference notes	599	16-ai-tech-stack/tracks/claude-code/code/claude-md-template.md 16-ai-tech-stack/tracks/claude-code/code/example-skill/README.md 16-ai-tech-stack/tracks/claude-code/code/example-skill/SKILL.md 16-ai-tech-stack/tracks/claude-code/code/hooks-README.md
16.7.6	Primary sources & further reading	612	16-ai-tech-stack/tracks/claude-code/papers/PAPERS.md 16-ai-tech-stack/tracks/claude-code/resources/RESOURCES.md
Part 99 — 🎓 Capstone Projects			
99.1	Overview	616	99-capstones/README.md
Appendices — Appendices			
A.1	How this book was built	619	generated
A.2	Chapter index	621	generated
A.3	Authoring Spec — AI End-to-End Learning Track	629	_tools/AUTHORING_SPEC.md
A.4	Glossary & cheatsheets	632	_shared/glossary/README.md _shared/cheatsheets/README.md